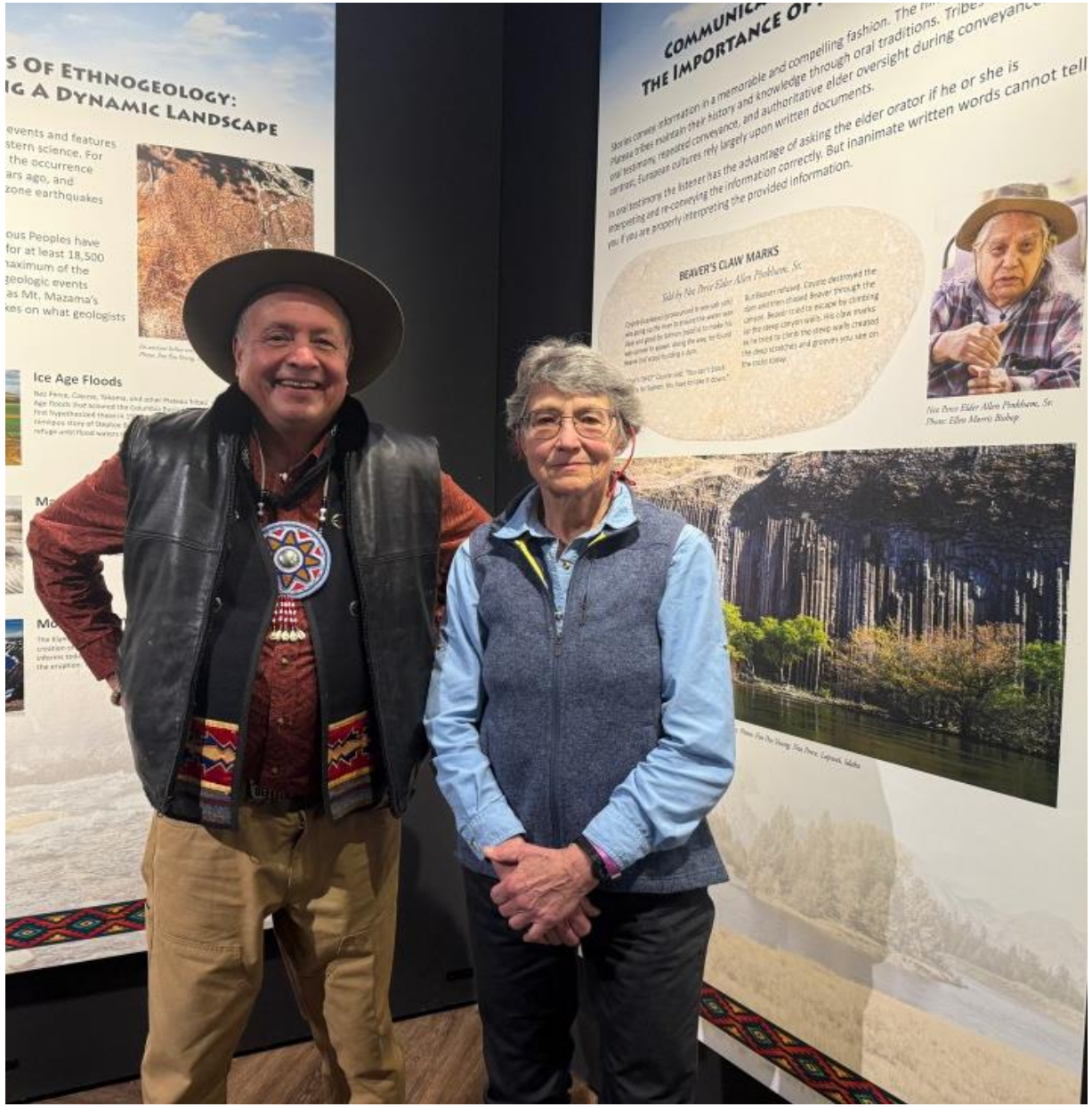
Presented by Fady Youssef

"Geology with a Soul": Indigenous Knowledge as Science



- Ethnogeologist Roger Amerman, Choctaw Nation citizen (left)
- geologist Ellen Morris Bishop (right)

- Both geologists practiced ethnogeology during their researching process.
- Used Nez Perce (Nimiipuu) legends as observational data to solve geological mysteries that Western science missed.

What is Ethnogeology?

Definition:

The scientific study of geological features through the lens of Indigenous cultural knowledge.

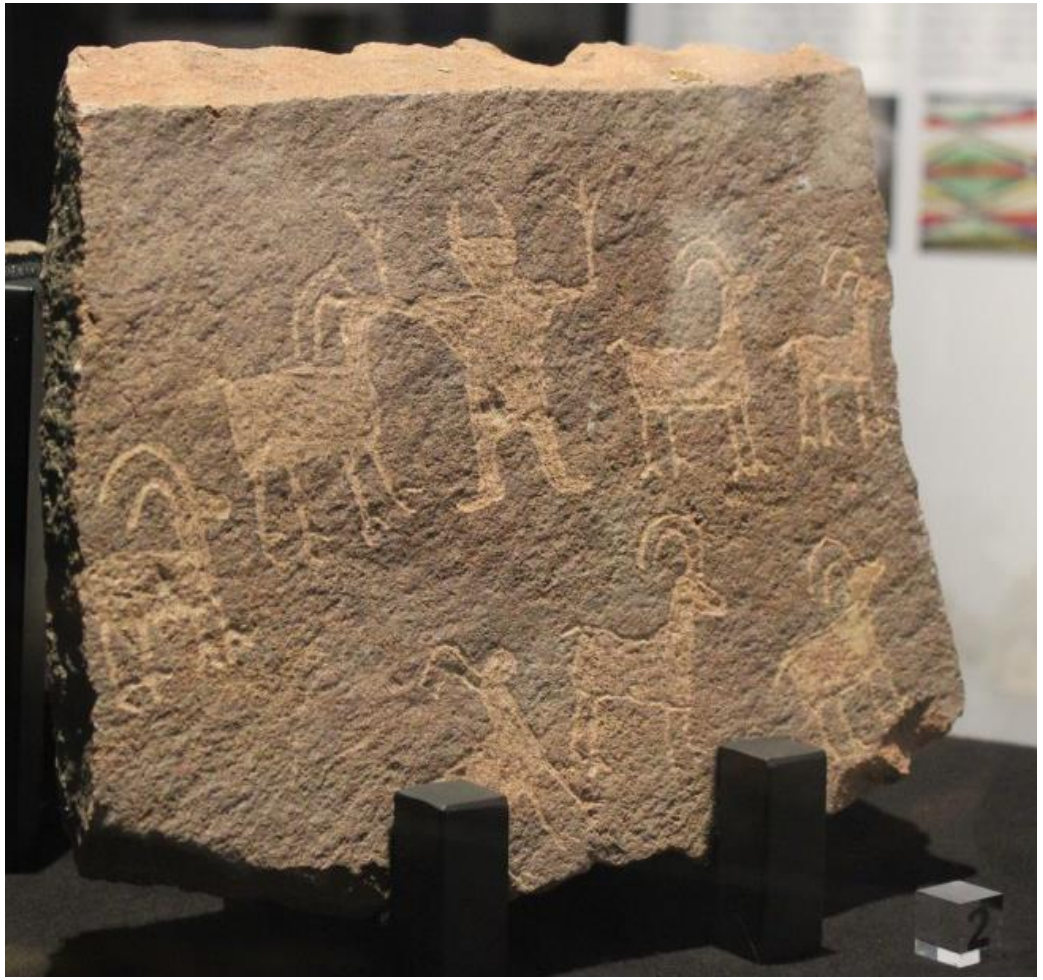
Approach to Data

Western Science: Physical examination of Earth's structure and composition to understand its past.

Ethnogeology: Oral traditions and cultural stories as observational records to interpret the landscape's history.

For Morris-Bishop and Amerman, these details gave them important insight into how far and high floodwaters extended.

“You need to be able to tell people about these floods, but you need characters in a story,” she said. “So it’s better if you have the mythical old man who climbs to the top of a rock in order to fish when there’s a big flood.”



One of the legends an elder provided were of historic floods coming through Hells Canyon around 15,000 years ago stating the Nez Perce climbed the top of Steptoe Butte and shared the area with wildlife including cougars, bears, otters and raccoons who were seeking shelter from the flood.

Klamath oral history describes a war between the spirits Llao and Skell. The battle began with the earth shaking violently as the spirits signaled their anger. Then, they threw red-hot rocks at each other and the sky turned dark with ash (the eruption) before the mountain finally collapsed into itself to imprison Llao forever.

Geologist Charlie Bacon original found no evidence of a pre-eruption earthquake, leading to doubts of the oral history. Researchers later discovered a hidden fault scarp north of the lake using advanced LIDAR technology confirming the Klamath story was an eyewitness record of the event 7,700 years ago.

Sources

Primary News Source:

- Aquino, L. (2026, January 28). *“Geology with a soul:” New OMSI exhibit highlights Indigenous storytelling and geology*. ICT News. https://ictnews.org

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